

Flick Typing: Toward A New XR Text Input System Based on 3D Gestures and Machine Learning

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ABSTRACT

We propose a new text entry input method in Extended Reality that we call Flick Typing. Flick Typing utilizes the user's knowledge of a QWERTY keyboard layout, but does not explicitly provide visualization of the keys, and is agnostic to user posture or keyboard position. To type with Flick Typing, users will move their controller to where they think the target key is with respect to the controller's starting position and orientation, often with a simple flick of their wrists. Machine learning model is trained and used to adapt to the user's mental map of the keys in 3D space.

Keywords: Extended Reality, Text Input, Gesture Mapping

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction techniques—Text input

1 INTRODUCTION

1.1 Background

Text input systems play a significant role in various Extended Reality (XR) fields. Currently, using a virtual representation of a standard QWERTY keyboard in some form is a common way to input alphabetical letters [3]. However, the virtual QWERTY keyboard used by such input systems usually takes up a large area of the user's field of view which may make user interaction inconvenient. Therefore, virtual keyboard may hinder user experience for XR applications.

1.2 3D Flick Typing System

We propose a new framework of alphabetical typing system called 3D Flick Typing which utilizes 3D gestures in space to uniquely input alphabetical keys. Our aim is to develop a typing system with friendly and personal customized user interaction experience, high accuracy, high speed and high degree of freedom. The mechanism is similar to the Punch Typing system proposed by Yao et al [7]. It is called Flick Typing since users can perform gestures by just flicking wrists and slightly moving hands position. To make the system generalizable and convenient to use, we apply machine learning technique to construct the mapping from gestures in 3d space to keyboard entries. When using our Flick Typing system, users can be in any body postures while typing, including sitting, standing, lying down, or even walking. It only requires users to have knowledge about the layout of standard QWERTY keyboard, and proficient users can even type without looking at any visual aid. We implemented our Flick Typing system in Unity 2019, and tested the system on HTC Vive 1st generation.

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2 RELATED WORK

We look at four common types of typing solutions for XR that have been explored in recent years: (1) Visualize hands and keyboard in VR scenes. These methods use cameras on the HMD to track the position and motion of user's hands and visualize them in VR scenes [6]. There would be a virtual keyboard in the scene with which users can interact. Users can type as if on the physical keyboard with high speed using these methods. (2) Design specific usage on hand controllers. These methods focus on making full use of two hand controllers, such as designing specific operation patterns for the buttons of hand controllers [8], or utilize different movements of controllers. [1]. (3) Using additional equipment. These methods use hardware equipment other than HMD and hand controllers to help typing [4]. (4) Eye tracking. Research has gone into use gaze-tracking technology to type words instead of using hands [5]. The keys are specified by the movement of user's eyes.

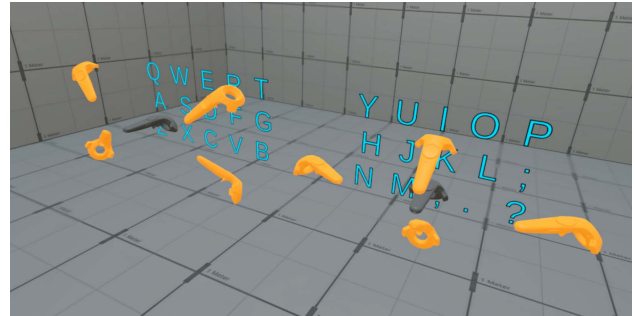


Figure 1: Keyboard layout and 3D gesture schema. Controllers initially point at 'D' and 'K' respectively shown by grey models. The 4 orange models at left show the ending status of gestures mapping to 'Q', 'T', 'Z' and 'B'. Other 4 orange models at right show the ending status of gestures mapping to 'I', 'H', ';' and ','.

3 3D FLICK TYPING FRAMEWORK

3.1 Input Mechanism

Flick Typing is based on the QWERTY layout. As shown in Figure 1, a simplified QWERTY keyboard containing 30 keys is divided to 2 parts, the left and right hand responsible for each part. The user's hand controller is pointing at the central key of the corresponding sub-keyboard, namely the left controller pointing at 'D' and the right controller pointing at 'K'. In order to input a letter, the user would use a button on the controller (e.g., the grip button on the side of the HTC Vive controller) to initiate a gesture. Using the starting key as the reference coordinate, the user would hold down the button and move the controller from the starting key to the target key. Upon reaching the target key, the user would release the button and the key will be inputted. The movement can be performed easily by slightly flicking and moving the user's wrists, and different keys are distinguished by different flicking direction and distance. For example, user should flick the right hand outside substantially and to around 2 o'clock to type 'P' (as shown in Figure 2), while move



Figure 2: Typing P in Different Orientations

the right hand slightly to around 1 o'clock with little motion of wrist to type 'O'. Users do not need to care where the key exactly is. The only need is to move the controller smoothly according to their spatial memory of the QWERTY keyboard.

3.2 Key Prediction Based on Machine Learning

Due to their strong capability for regression and classification, We use Multi-layer Perceptron Neural Networks (MLP) to construct the mapping between hand gestures and keys [2], one classifier for each hand. We propose a set of input data features for the MLP models using the local transformation of the hand controllers with respect to their start status during a gesture in order to avoid the influence of the orientation of the body or the position of the hands. The 7 input features for each hand are as follow:

Local position measured by Cartesian coordinates (x, y, z)

Local rotation measured by Quaternions (x, y, z, w)

The output of the MLP network is 15 confidence scores representing the probability that the gesture represents the key's index.

4 EXPERIMENTS AND PRELIMINARY RESULTS

4.1 Machine Learning Results

We set up three experiments to test the performance of our system:

(1) Train a universal model using one part of all of four participants' training data and test on the other part of these user's data. The average prediction accuracy reaches 91.60%.

(2) Train user-customized models for each participant. Since it is not practical for an input system to require the users to provide a lot of training samples before using the system, we perform experiments on testing whether our system can work well with a small quantity of training samples. We train models using 10/20/40 samples per key from one participant, results recorded in Table 1.

(3) Train a model using data collected from different body postures, including facing front, facing right and lying down. 5 samples per key are collected for each posture from one user, and the test accuracy reaches 94.45%.

Table 1: Machine Learning Accuracy

Num of Samples / Key	Left Hand	Right Hand
10	79.17%	76.87%
20	84.97%	85.33%
40	95.77%	94.07%

4.2 Typing Speed Test

We test the typing speed measured by word-per-minute (WPM) value. One word is defined to be 5 characters long. We perform 2 sets of experiments: one is to type with highest possible accuracy, the other is to type with highest possible speed. For the former experiments, the user is required to type with 100% accuracy with the help of a

real-time display mechanism, which shows the prediction result as soon as the user start to perform gestures. The typing speed finally reaches an average of 8.77 WPM. For the latter experiments, the user is asked to type as fast as possible with little attention on the keyboard and real-time display. The typing speed reaches an average of 15.16 WPM, with an average accuracy of 84.29%.

5 CONCLUSION

Flick Typing is currently a work in progress. However, it has demonstrated its potential in text input work in XR applications. With simple flicks of wrists according to spatial memory of QWERTY keyboard layout and the real-time display mechanism, users can type words with relatively high speed and accuracy. We have proved that users can type without limitation on where they stay, which direction they face, whether they are standing or lying. In order to make our typing system generalizable and user-customized, we apply neural networks for building the mapping from spatial gestures to keys. According to the experiments we perform, the system can reach a relatively high accuracy with small amount of training samples. We will keep optimizing the Flick Typing system and perform more abundant experiments, making it a new typing methods practical in industrial and commercial XR applications.

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